



NIL-UCM at the CLEF 2026

SimpleText Track:

Prompting, Fine-Tuning, and Multi-Agent
Architectures for Biomedical Text Simplification

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Develop and evaluate LLM-based systems for automatic scientific text simplification, comparing three methodological strategies:

1 Prompting

Analyse strategies of increasing complexity at sentence and document level, and assess 4-bit quantization as a viable alternative.

2 Fine-tuning

Evaluate whether supervised fine-tuning with LoRA improves over prompting in a low-resource biomedical setting.

3 Multi-agent architecture

Design a system where specialised models collaborate to produce simplifications that are both factually accurate and linguistically accessible.

Experimental framework



SimpleText @ CLEF 2026

Task focused on the automatic simplification of scientific texts

Task 1.1 — Sentence-level

Given a single technical sentence →
generate a simplified version

Task 1.2 — Document-level

Given a complete technical document →
generate a **Plain Language Summary (PLS)**



Corpus:
Cochrane-auto

- **Cochrane systematic review documents** and their **PLS**
- **Main 2026 test set:** 175 documents (~3,700 sentences) with no access to the PLS
- **Public corpus:** training, validation, and test sets from previous editions

PROMPTING

MODELS



Llama 3.1 8B Instruct



Mistral 7B Instruct v0.3



Gemma 2 9B Instruct



4 bits quantization
versions (Llama & Mistral)

STRATEGIES

Task 1.1 (Sentences)

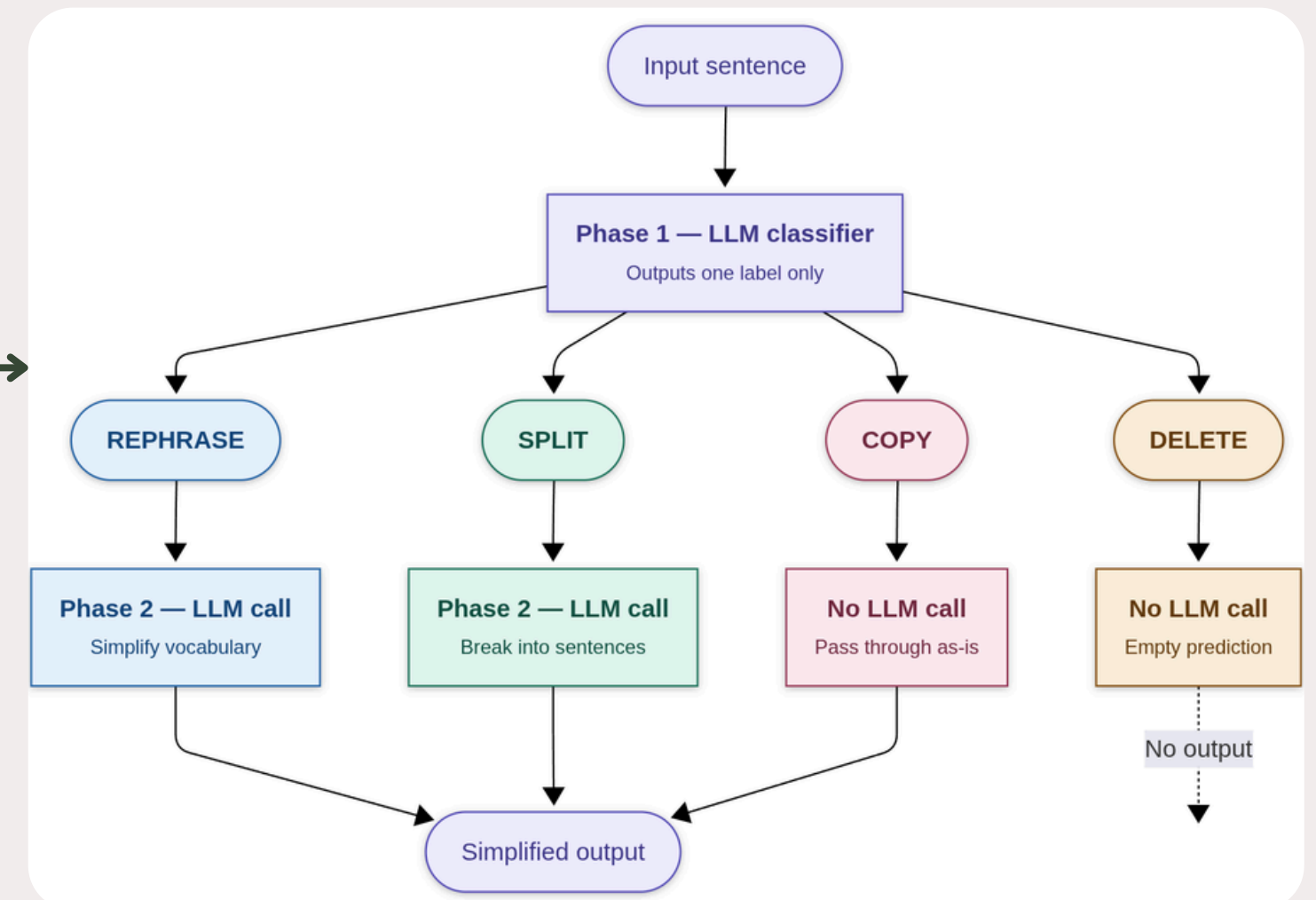
- **Role-based**

Role of a medical writer specialised in PL simplification

- **Few-shot**

Real examples from
the public corpus

- **Two-phase**



PROMPTING

MODELS



Llama 3.1 8B Instruct



Mistral 7B Instruct v0.3



Gemma 2 9B Instruct



**4 bits quantization
versions (Llama & Mistral)**

STRATEGIES

→ Task 1.2 (Documents)

- **Role-based and Few-shot**
- **Reference-informed**
Common patterns in Cochrane references
- **Guidelines-based**
Cochrane Plain Language guidelines
- **Guidelines + Glossary**
Automatically adds definitions of identified medical terms

FINE-TUNING

MODELS



Llama 3.1 8B Instruct



Mistral 7B Instruct v0.3

LoRA TECHNIQUE

- **Parameter-efficient training**
- Implemented with **Unsloth** and trained on **Google Colab Pro** (A100)

DATA

- **Source:** public Cochrane-auto corpus
- **Training:** 849 technical document – PLS pairs
- **Validation:** 119 pairs

CONFIGURATION

- **Prompt:** Role-based
- **Maximum length:** 4.096 tokens

Methodology



MULTI-AGENT SYSTEM

MODELS

 Llama 3.3 70B Versatile

 Mistral Small 4

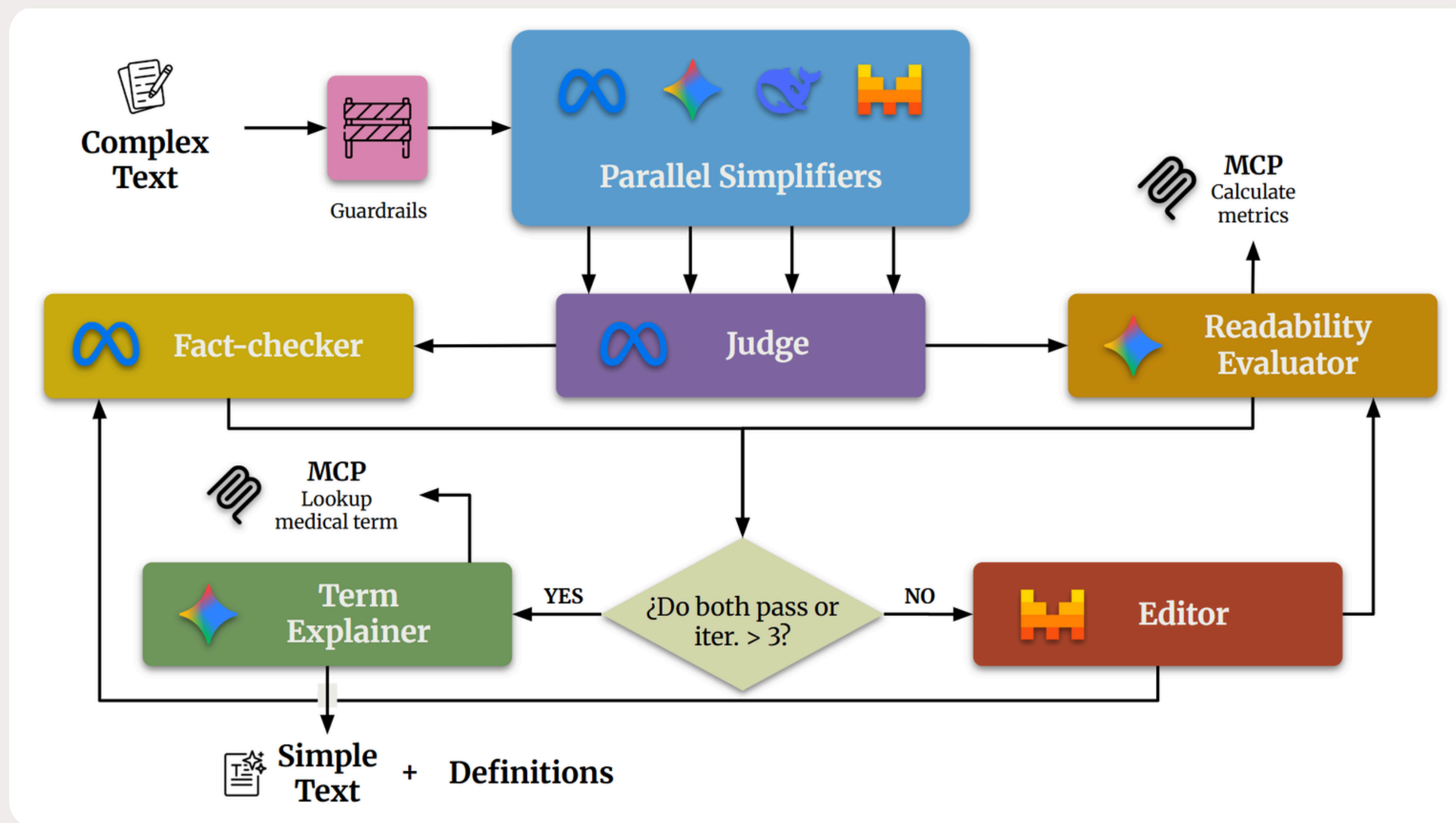
 Gemini 2.5 Flash Lite

 Deepseek-V4 Flash

Orchestration:
LangGraph 

External tools:
MCP 

Web application:
Streamlit 





TASK 1.1

Sentence-level simplification



Role-based strategy
54th place



Two-phase strategy
Models fail at classifying the operation



4-bit quantization:
hardly affects SARI

	PROMPTING STRATEGIES		
	Role-based	Few-shot	Two-phase
Mistral	44,87		40,16
Mistral Q	44,33		
Llama	45,51	45,43	42,25
Llama Q	45,38		
Gemma	44,91		44,40

Results and discussion



TASK 1.2

Document-level simplification

MULTI-AGENT SYSTEM	
Role-based	Guidelines + Glossary
45,34	44,89

	PROMPTING STRATEGIES					FINE-TUNING
	Role-based	Few-shot	Reference-informed	Guidelines-based	Guidelines + Glossary	
Mistral	43,84		42,01	41,43		35,95
Mistral Q	42,59					
Llama	44,40	42,58	44,18	44,99	44,91	32,54
Llama Q	44,34					
Gemma	43,29			43,98		



Multi-agent System
24th place



Fine-tuning
Models learn to copy fragments



4-bit quantization:
hardly affects SARI

1 Prompting

- Competitive simplifications with no training at all
- 4-bit quantization works with practically no loss in quality

2 Fine-tuning

- Underperforms with the available data
- Models learn to copy rather than simplify

3 Multi-agent architecture

- Most competitive approach: 24th place on Task 1.2
- Splitting the task into specialized roles is a good strategy

FUTURE WORK

- Explore strategies from other participants: top SARI scores were 47.43 (sentence-level) and 48.85 (document-level)
- Run a human evaluation: SARI doesn't capture naturalness or real usefulness for non-expert readers
- Explore more complex multi-agent configurations
- Align prompts with the actual stylistic patterns of Cochrane PLS, not just the editorial guidelines



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